Noise sensitivity and collision histories in the one-dimensional balloon process

Proof sketches

Working exposition prepared for Jacob Richey

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Abstract

Start equally growing balloons at an odd number of independent uniform points of a circle, and remove both balloons whenever they meet. Moving one starting point can change the final survivor and a chain of earlier collisions. We present proof sketches for two descriptions of this effect. A single resampling changes the survivor with a nontrivial limiting probability; every growing number of resamplings changes it with probability tending to one. For $1 \le k = o(N)$ resamplings, the number of changed collision pairs is asymptotic in L^1 to $(1 + e^{-\gamma})k \log(N/k)$. A moving discrepancy connects the two results. Recording its origin in a tree of gap mergers gives the survivor response, while summing its captures between the input and edit scales gives the history cost. The final section on fluctuations explains how the same construction leads to Gaussian limits and to a Poisson law for rare shared edges near complete resampling.

Scope. This AI-written exposition presents the working arguments in [W], through Section 7.61. The finite coupling and shielding lemmas receive proofs; the limit arguments receive sketches with references to the full record. These asymptotic results retain their working-proof status, pending independent verification. The value near 0.658 remains an uncertified estimate.

1 Introduction

Place an odd number of points uniformly on a circle. Grow a balloon from each point, removing two balloons as soon as they touch, until only one remains. Now move one starting point and repeat the construction. The survivor changes with a probability bounded away from zero, even as the number of points grows. The effect also appears in the earlier collisions: typically, order $\log N$ pairs change. We ask how this response grows when several starting points are moved, and what connects the surviving balloon to the history that produced it.

Let N be odd and let $X_1, \dots, X_N$ be independent uniform points of the circle $\mathbb{R}/\mathbb{Z}$, of circumference one. Each balloon grows at unit speed. The process equivalently matches and removes the closest remaining pair at each step. Distances are intrinsic circle distances; ties have probability zero. We use the separation r of a colliding pair as the time parameter, so physical time is $r/2$.

Write $C_r(A)$ for the active points of an input A after all collisions of separation at most r, and $\mathcal{M}_r(A)$ for the edges accepted by then. The completed matching is $\mathcal{M}(A) = \mathcal{M}_{1/2}(A)$. Each odd input has $(N-1)/2$ edges and one survivor.

Take an independent uniform sample $Y_1, \dots, Y_N$ and define the nested resampling path

$$A_k = \{Y_1, \dots, Y_k, X_{k+1}, \dots, X_N\}, \qquad 0 \le k \le N.$$

The point in coordinate i keeps label i when its location is replaced. Let $p_N(k)$ be the probability that A_0 and A_k have different survivor labels. Define the spatial history distance by

$$H_N(k;r) = |\mathcal{M}_r(A_0) \triangle \mathcal{M}_r(A_k)|, \qquad H_N(k) = |\mathcal{M}(A_0) \triangle \mathcal{M}(A_k)|.$$

Thus a pair removed from one matching and a new pair added to the other contribute two to $H_N(k)$. Edges here are identified by their physical endpoints. Identifying them by coordinate labels instead gives $H_N^{\mathrm{lab}}(k)$. The history theorem below applies to both conventions; their Poisson laws near complete resampling are described separately in Section 7.3.

Theorem 1.1 (Survivor noise)**.** *There is a constant $p_\infty \in (0,1)$ such that $p_N(1) \to p_\infty$. For every fixed k, $p_N(k) \to p(k)$, where*

$$1 - p(k) = \int_0^1 (1-x)^k \, \nu(dx) \tag{1.1}$$

for a probability measure ν on $[0,1]$ satisfying $\int x \, \nu(dx) = p_\infty$ and $\nu(\{0\}) = 0$. Consequently, for every sequence $k_N \to \infty$ with $k_N \le N$,

$$p_N(k_N) \longrightarrow 1.$$

The fixed-k limits also describe disagreement of survivor positions.

The assertion for growing k_N is stronger than existence of the one-point limit. It rules out any part of the survivor agreement that would persist under arbitrarily many resamplings. Under independent resampling with probability s_N per coordinate, the transition occurs when the expected number Ns_N of replacements is of order one. If $Ns_N \to 0$ there is usually no replacement; if $Ns_N \to \infty$ the survivor labels disagree with probability tending to one. At $s_N = \lambda/N$ the limiting disagreement is $1 - \int e^{-\lambda x} \nu(dx)$.

Theorem 1.2 (The size of the changed history)**.** *Let γ be Euler's constant. For every nonzero sequence $k = o(N)$, including fixed k,*

$$\frac{H_N(k)}{k \log(N/k)} \longrightarrow 1 + e^{-\gamma} \qquad \text{in } L^1. \tag{1.2}$$

Moreover, uniformly over $1 \le k \le N$,

$$\mathbb{E}H_N(k) \asymp k(1 + \log(N/k)), \qquad \sup_{1 \le k \le N} \mathbb{E}\left| \frac{H_N(k)}{\mathbb{E}H_N(k)} - 1 \right| \longrightarrow 0.$$

The same statements hold for H_N^{lab}.

For $k \to \infty$ with $k = o(N)$, these theorems give a different survivor with high probability and $H_N(k)/N \to 0$. Almost every collision pair is preserved. The changed pairs form paths through that matching, and their total length measures how the perturbation reaches the survivor.

Proof strategy

The basic comparison is an insertion into a common background. Adding one point creates exactly one active discrepancy between the two systems. When it collides with a common point, the discrepancy passes to that point in the other system. Thus its path records both the lost and the new collision edges. A resampling is a comparison of two insertions into the same even background.

To determine the survivor, we must keep track of where this discrepancy ends. In one dimension, each collision merges three consecutive gaps. Reversing those mergers gives a tree. We mark each merger with the map that transports insertion origins, and show that the expected mass left undetermined below depth d decays geometrically in d. This estimate allows finite trees to determine the limiting insertion map. It also constructs a joint *entrance law*: the coupled processes obtained as the initial sample grows, with finitely many active points at each positive separation and with shared point identities retained.

The insertion map partitions origins according to their final survivor. Its basin masses give the one-resampling agreement probability. Coordinate orthogonality then expresses the full noise profile through a probability measure ν. The remaining step is to exclude an atom of ν at zero. We do this by targeting a hypothetical persistent survivor with an insertion near its location, while preserving the stationary law of the entrance history.

For the history cost, we count the collisions reached along a discrepancy path. The expected number of captures per logarithmic scale has a limit determined by the marked gap law. With k edits, these contributions accumulate between spacings $1/N$ and $1/k$; interactions after that range cost only order k in mean. This gives the factor $\log(N/k)$ and the coefficient in Theorem 1.2. Concentration uses averaging over insertion origins for growing edit counts and over logarithmic scales for fixed counts. The fluctuation results require additional bounds on the centered interaction cost.

The proof brings together the matching and coarsening descriptions of the dynamics. Angel, Ray and Spinka [1] study balloons on infinite spaces, and Holroyd, Pemantle, Peres and Schramm [2] develop stable matchings of point processes. The one-dimensional triple-gap merger rule, cyclic exchangeability, and a related retained-mass recursion appear in Bray, Derrida and Godreche [3]. Faggionato, Roberto and Toninelli [4] develop renewal scaling limits for hierarchical coalescence. Here the origin marks connect the gap evolution to the response of the matching under resampling.

The perturbation question also has a close mean-field counterpart. Ahlberg, Deijfen and Sfragara [5] prove robustness of most matching edges and sensitivity of late edges under resampling of independent edge costs. Here the randomness lies in point locations, so the edge distances are strongly dependent. The marked gaps and their coupled entrance laws handle this spatial dependence.

2 A moving discrepancy

Write $B + z = B \cup \{z\}$. To compare B and $B + z$, scan all pairs in their union in increasing order of distance. Each copy accepts a pair when both endpoints are active in that copy. A point active in only one copy is a *discrepancy*. The following observation controls every later comparison.

Lemma 2.1 (Discrepancy contraction)**.** *Couple two finite inputs A, B using one ordered pair-event schedule, with a common rule for ties. For $s \geq r \geq 0$,*

$$|C_s(A) \triangle C_s(B)| \leq |C_r(A) \triangle C_r(B)| \leq |A \triangle B|. \tag{2.1}$$

If $B = A + z$ with $z \notin A$, the discrepancy count is exactly one at every cutoff.

Proof. An event accepted in both copies removes two common points and preserves the discrepancy count. An event accepted in just one copy contains at least one discrepancy. If its other endpoint is common, that endpoint becomes the discrepancy in the other copy, preserving the count. If both endpoints are discrepancies, the count decreases by two. All other events leave both configurations unchanged. Thus the count is nonincreasing and keeps its parity. A count initially equal to one remains equal to one. □

For a resampling, the initial discrepancy count is two. Its changed matching edges trace the successive transfers of these discrepancies. Several edits give several paths, which may meet and close; each accepted event either transfers a discrepancy or removes a pair of them.

The useful feature in one dimension is that the path can be encoded without following the inserted process separately. Consider the next background collision, between u and v, with separation s, and let a, c be the adjacent gaps. In local coordinates put $u = 0$, $v = s$ and

$$\alpha = \min(s, a/2), \qquad \beta = \min(s, c/2).$$

An insertion at $z \in (-\alpha, s/2)$ kills u before the background collision, leaving v as the discrepancy. The symmetric event leaves u. Accordingly define

$$T(z) = \begin{cases} s, & -\alpha < z < s/2, \\ 0, & s/2 < z < s + \beta, \\ z, & \text{otherwise.} \end{cases} \tag{2.2}$$

If $S_B(z)$ denotes the survivor position after inserting z into an even background B, the exact recursion is

$$S_B(z) = S_{B\setminus\{u,v\}}(T(z)). \tag{2.3}$$

The map gives the discrepancy's location after this background pair has gone. It transports the origin's effect through the collision history.

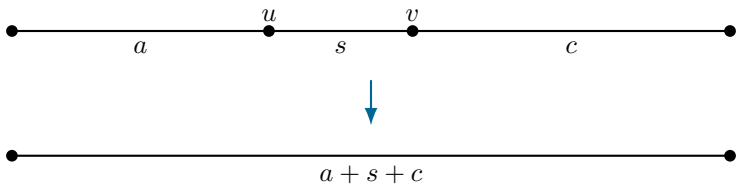


Figure 1: A collision merges three gaps. The reversed mergers form a tree, and the collapse map at each vertex records how insertion origins pass through it.

Compose the maps in collision order and push Lebesgue measure through them. At an intermediate cutoff, each active gap carries a measure of total mass equal to its length. Its atoms record origins whose discrepancies have already been transferred to old collision vertices. After the even background is exhausted, this gives

$$(S_B)_{\#}\,\mathrm{Leb} = \mathbf{1}_{E_B}(z)\,dz + \sum_{v \in B} \theta_v(B)\delta_v,$$

where $E_B = \{z : S_B(z) = z\}$ and $\theta_v(B) = \mathrm{Leb}\{z : S_B(z) = v\}$. Thus θ_v is the basin mass of the background label v. Let $\Theta_N = \mathrm{Leb}(E_B)$, with $N = |B| + 1$. Two independent insertions into the same background agree in a background label with conditional probability $\sum_v \theta_v^2$; they agree in the insertion label with probability Θ_N^2. Consequently

$$p_N(1) = 1 - \mathbb{E}\sum_v \theta_v^2 - \mathbb{E}\Theta_N^2. \tag{2.4}$$

Exchangeability gives $\mathbb{E}\Theta_N = 1/N$, so the last term vanishes.

The noise limit now reduces to convergence of the basin masses and their squared sum. This is a stronger requirement than weak convergence of the active configurations: the readout depends on the identities of the atoms carrying the insertion origins. We therefore retain the collision history inside each gap and control the mass that remains unresolved when that history is truncated.

Full argument: [W, Sections 2, 7.1, and 7.9].

3 Marked entrance laws

3.1 Renewal gaps and their merger trees

The local ingredient is an interval whose endpoints survive while its interior points match among themselves. We call this event *shielding*. It allows the past inside a current gap to be treated as a separate random object.

Lemma 3.1 (Interval shielding)**.** *Put k independent uniform points in an interval of length $x > 0$, keep balloons at its two endpoints, and ignore their mutual collision. The probability that all interior points disappear while both endpoints survive is zero for odd k and $1/(k+1)$ for even k. On this event every interior matching distance is at most $x/3$.*

Proof. Interior pairs remove an even number of points. For even k, identify the endpoints to obtain a circle of circumference x with a distinguished point. A pair using the shortcut through that point has an endpoint closer to the distinguished point than to its proposed partner. Hence, while the distinguished point survives, all accepted pairs agree with the interval process. This identifies shielding with its survival.

Rotate $k+1$ iid uniform circle points so that the distinguished one is at zero. The other points remain iid uniform, and exchangeability makes each of the $k+1$ labels equally likely to survive. The probability is therefore $1/(k+1)$. Before each collision of this odd circle process, at least three points remain. Their smallest cyclic gap is at most $x/3$, so the next matching distance is at most $x/3$. □

At separation r, every active gap has length at least r. On the line, the entrance law is a stationary renewal process whose gap density is

$$w_r(x) = r^{-1} f(x/r), \qquad \int_1^\infty e^{-sy} f(y)\, dy = \tanh\left(\frac{E_1(s)}{2}\right), \qquad E_1(s) = \int_s^\infty \frac{e^{-t}}{t}\, dt. \tag{3.1}$$

The triple-merger recursion gives an equation for the Laplace transform $F(s) = \int e^{-sy} f(y)\, dy$:

$$sF'(s) = -\tfrac{1}{2} e^{-s}(1 - F(s)^2), \qquad F(\infty) = 0.$$

Integrating the equation for $\operatorname{arctanh} F$ gives (3.1). Expanding at zero gives mean gap $2e^\gamma r$, so the intensity is c/r, where

$$c = (2e^\gamma)^{-1}. \tag{3.2}$$

On a circle the gaps obey a cyclic renewal law, conditioned to have total length one and the prescribed count parity. Conditional on this gap geometry, the past trees inside different gaps are independent.

To describe one tree, protect the endpoints of an interval of length x and suppose its interior has cleared by separation $t \le x$. The last interior collision divides the interval into lengths (a, s, b), where $b = x - a - s$ and s is the separation of that collision. The reverse split has density

$$\mathcal{K}_{t,x}(a, s) = \frac{w_s(a) w_s(s) w_s(b)}{w_t(x)}, \qquad 0 < s < \min(t, x/3), \quad s < a < x - 2s. \tag{3.3}$$

After the split, the three daughter trees are independent, with inherited cutoff s. The factors in the numerator describe their independent clearances; the denominator normalizes the conditioning on the parent interval. Iterating gives the full merger tree, and attaching (2.2) at each merger gives its origin measure.

There are two distinct convergence steps here. At any fixed cutoff and fixed tree depth, one first proves convergence of the finite geometry in total variation. The finite densities can be written as coefficients of products of Poisson interval weights. Expanding those weights isolates the terms consistent with the total count and parity; the remaining terms vanish. Nonnegativity and convergence of total mass then upgrade density convergence to total variation. This is a finite-depth calculation. Taking infinite depth requires a separate estimate.

The estimate uses a positive representation of f as a geometrically weighted convolution sum of another probability density. Exponential tilting turns it into a renewal density and gives constants $b_-, b_+, \eta > 0$ such that

$$b_- e^{-\eta y} \le f(y) \le b_+ e^{-\eta y}, \qquad y \ge 1. \tag{3.4}$$

The lower bound matters as much as the upper bound. In (3.3), it gives a uniformly positive probability of a balanced split, even after conditioning on the parent state.

3.2 Cumulative queries and origin maps

Suppose we want the mass $\mu([0,z])$ in a shielded interval. Reveal the last split (a,s,b) and express this quantity in terms of the daughter cumulative masses. Outside the middle daughter, only one daughter remains unresolved. Inside it, the two collapse cells must be read simultaneously. With F_1, F_2, F_3 the daughter cumulative masses and $\alpha = \min(s, a/2)$, $\beta = \min(s, b/2)$, for example,

$$\mu([0,z]) = F_1(a-\alpha) + s - F_2(s/2) + F_3(\beta), \qquad a < z < a+s.$$

Thus a cumulative query follows one or three daughter queries, each with coefficient of absolute value one. A query at an endpoint is already known from the total mass.

Give each unresolved terminal interval a budget equal to its length. For any two possible completions below depth d, the difference in the queried masses is at most the total budget Z_d. This is a deterministic bound once the first d levels are known. For every parent state and query threshold, (3.4) supplies an event of uniformly positive probability on which the query follows one side daughter of length at most $3x/4$. For every other split the retained budget is at most x. Iterating this conditional contraction gives, for some $\rho < 1$,

$$\mathbb{E} Z_d(t,x,z) \le x\rho^d. \tag{3.5}$$

For several queries, we sum their error budgets. The same estimate applied on both sides of a threshold, together with continuous finite-tree split laws, excludes mass at the readout boundaries.

At a fixed positive cutoff there are only finitely many active gaps and finitely many future collision maps. The final basin masses of the active labels can therefore be read using finitely many such queries. Couple the finite and limiting depth-d geometries, apply the same deterministic budgets to their deeper completions, and then let $d \to \infty$. This gives convergence of the relevant mass vector, and hence of its squared norm. Finally send the cutoff to zero. An early-label estimate shows that labels which have disappeared before that cutoff make a vanishing contribution to the agreement probability. The order of limits is essential: sample size, then tree depth, then the positive cutoff.

Coupling insertions requires the image of the *same origin* in the two trees. Follow a uniform origin backwards through its unique daughter interval. Once it enters a middle daughter, a collapse assigns it to a collision vertex. A killed ancestry estimate shows that the mass of origins which avoid all middle daughters decays with depth. Combining this with the threshold-query bound determines which vertex is chosen. The resulting maps $\Phi^{(d)}$ satisfy, for some $\sigma < 1$,

$$\mathbb{E}\,\mathrm{Leb}\{z : \Phi^{(d)}(z) \ne \Phi(z)\} \le Cx(x/t)\sigma^d. \tag{3.6}$$

Equality here means the same vertex. Almost every origin eventually has an exactly determined image, and the limiting origin measure is purely atomic on the countable set of collision vertices.

3.3 Several insertions and shared identities

Represent a comparison with k resamplings as two sets of k fresh probes inserted into a common background of size $N-k$. More generally, take q probes and any fixed family of subsets of them. Stop the background at a small cutoff ε. If all probes lie in distinct, nonadjacent active gaps, their responses superpose exactly at that cutoff, simultaneously for every subset. Their images either add a point or remove the nearest active endpoint, according to the one-insertion rule.

If the active gap lengths are x_i, a union bound controls failure by the empty-background probability and a constant times $q^2 \sum_i x_i^2$. The renewal estimates give

$$\limsup_{N\to\infty} \mathbb{P}(\text{failure at } \varepsilon) \le Cq^2\varepsilon \tag{3.7}$$

for fixed q. On the good event, the convergent marked gap trees determine a unique joint law. Push this law to any later positive cutoff. Every subsequential joint limit is within the error in (3.7) of that same construction. Sending $\varepsilon \downarrow 0$ proves uniqueness.

The marks specify the set of copies containing each background point. The separated construction also rules out accidental coincidence of distinct limiting identities. Survivor agreement is therefore a valid readout of the joint limit, because the construction keeps precisely the identities tested by that event.

Random deletions have the same limiting response as fresh insertions. An insertion at the location of an existing point immediately cancels that point and realizes its deletion. The empirical measure of the original sample and Lebesgue measure have asymptotically equal masses on all intervals. At any fixed tree depth this compares their origin budgets, including intervals chosen from the sample itself. The depth error then vanishes by (3.6). This is the reason a resampling can eventually be represented by two insertions in the entrance process.

Full argument: [W, Sections 2, 7.4–7.6, 7.10, and 7.13–7.14].

4 Survivor noise

4.1 A nontrivial limit and its spectrum

The marked entrance law makes the basin identity (2.4) convergent, giving p_∞. The geometry of the last background pair then bounds this limit away from zero and one. For $N \ge 3$, let L be the shorter separation of that pair in the even background of size $N-1$. Its two arcs clear internally without touching their endpoints. Conditional on the last endpoints and the allocations to the arcs, the two interior configurations are independent and reflection-symmetric. A uniform insertion in the shorter arc leaves one of the two endpoints as survivor, with an averaged fair choice between them. An insertion in the other arc can agree with at most one of those choices. The fair choice is between two old endpoint labels, so the argument bounds label disagreement directly. Thus

$$p_N(1) \ge \mathbb{E}[L(1-L)].$$

On $1/4 \le r \le 1/2$ the last-pair density for an even background of size M is explicitly

$$\frac{1-(1-2r)^M}{2r(1-r)}.$$

Integrating just this part gives $\liminf p_N(1) \geq 1/8$. Conversely, on $L > 1/3$ every insertion leaves one of the two last endpoints. Conditional disagreement is then at most $1/2$, yielding $\limsup p_N(1) \leq 1 - (\log 2)/4 < 1$. Sharper tree integrals and certified numerical evaluation improve these elementary bounds to

$$0.2016982228 \leq p_\infty \leq 0.8112015673. \tag{4.1}$$

The proposed value near 0.658 is still uncertified.

For the full resampling profile, encode the survivor label as a unit vector $F_N \in \mathbb{R}^N$. Decompose this function of the independent coordinates into orthogonal components indexed by subsets of coordinates. Let $w_{N,j}$ be the total squared norm of components using exactly j coordinates. Their nonnegativity gives

$$1 - p_N(k) = \sum_{j=0}^{N} w_{N,j} \frac{\binom{N-k}{j}}{\binom{N}{j}}, \qquad \sum_j w_{N,j} = 1. \tag{4.2}$$

Indeed a component contributes to agreement exactly when none of its coordinates is refreshed. Averaging the refreshed set of size k gives the displayed avoidance probability. Coordinate exchangeability makes this equivalent to replacing the first k coordinates.

Place mass $w_{N,j}$ at j/N to obtain ν_N. For fixed k, replacing the avoidance factor by $(1 - j/N)^k$ costs at most $\binom{k}{2}/N$. The coupled limit gives convergence of $p_N(k)$ for every fixed k, and these moments determine a unique measure ν on $[0, 1]$. This proves (1.1). Formula (4.2) also makes $p_N(k)$ increasing in k.

To obtain sensitivity for growing resampling counts, we must show that $\nu(\{0\}) = 0$. The possible obstruction is spectral mass at orders $j = o(N)$: those orders approach zero after rescaling even though $w_{N,0} = 1/N$. The next argument excludes that accumulation.

4.2 Removing an invariant atom

The right state for this step is the full entrance history $\omega = (C_\omega(r) : r > 0)$, recorded at rational cutoffs. It contains the merger trees and hence all the origin maps. Let Π_{e} and Π_{o} be its even and odd laws. Inserting at origin z gives a measurable transformation $\mathcal{T}_z$ which exchanges these parities. The same-origin construction and deletion comparison give

$$\mathcal{T}_z^2 = I, \qquad \mathcal{T}_z \mathcal{T}_w = \mathcal{T}_w \mathcal{T}_z, \tag{4.3}$$

almost everywhere, and each $\mathcal{T}_z$ preserves the equal mixture $\Pi = (\Pi_{\mathrm{e}} + \Pi_{\mathrm{o}})/2$ for almost every z.

These identities come from the joint finite constructions. Reversing an iid add-one pair is a uniform deletion from its larger member. Keeping the tagged origin in this reversal identifies the inverse with the same $\mathcal{T}_z$. Inserting two independent probes in either order identifies the two compositions. The empirical-origin comparison from Section 3 is needed here; convergence of unmarked configurations would not justify either identity.

Let $Kf(\omega) = \int_{\mathbb{R}} / \mathbb{Z} f(\mathcal{T}_z \omega)\, dz$. This is a self-adjoint Markov operator between the parity sectors. On odd histories, one refresh has kernel $Q = K^2$, a positive self-adjoint contraction. Let $\mathcal{I}$ be its invariant sigma-field, and let Λ_ω be the conditional law of the survivor location $U(\omega)$ given $\mathcal{I}$. The spectral interpretation of long refresh times gives

$$\nu(\{0\}) = \mathbb{E} \sum_x \Lambda_\omega(\{x\})^2. \tag{4.4}$$

One way to obtain this identity is to partition the circle into dyadic intervals. For each interval, the long-time limit of its survivor indicator correlation is the squared norm of its invariant projection. Refining the partition leaves exactly the diagonal atomic mass.

We can now target a hypothetical atom. Suppose X is an $\mathcal{I}$-measurable location. Its invariant representative extends to even histories so that $X(\mathcal{T}_z\omega) = X(\omega)$ almost everywhere. To see this extension, apply K to $e^{2\pi iX}$ and use equality in the L^2 contraction: the average still has modulus one, so the values being averaged agree.

Starting from an odd stationary history, choose the insertion origin z uniformly in a small interval of radius h about $X(\omega)$. This adaptive choice still gives an exactly stationary even history. Indeed, writing a_h for the uniform density on that interval centered at zero, measure preservation and invariance of X imply

$$\begin{aligned}\int\int a_h(z - X(\omega))\, f(\mathcal{T}_z\omega)\, dz\, \Pi_{\mathrm{o}}(d\omega) \\ = \int\int a_h(z - X(\omega'))f(\omega')\, dz\, \Pi_{\mathrm{e}}(d\omega') = \Pi_{\mathrm{e}}(f).\end{aligned}$$

For each fixed h, the sampling density is bounded, so the almost-everywhere identities remain applicable.

The geometric input must apply to this history-dependent target. On one full-measure event, endpoint removal holds simultaneously at every positive rational cutoff and every target location. Its proof follows the endpoint threshold backwards through a leftmost or rightmost tree branch. A uniformly positive chance of settling the query, together with control of the inherited cutoff, gives a neighborhood of successful origins. The vertex set $\bigcup_{r\in\mathbb{Q}_{>0}} C_\omega(r)$ is countable, and each of its vertices is protected at an earlier cutoff. Insertion creates no vertex outside that set. These facts give the simultaneous statement needed to substitute $X(\omega)$.

Thus, after the adaptive insertion, X is absent at any fixed cutoff r with probability tending to one as $h \downarrow 0$. Make one more, independent uniform insertion. This restores the odd stationary law and preserves X. For X to be the resulting survivor, it must either have survived the adaptive removal or be restored from the insertion mass carried by vertices already absent at cutoff r. If B_r is that early basin mass in the intermediate even history,

$$\mathbb{P}(U = X) \le \mathbb{P}\{X \text{ remains active at } r \text{ after targeting}\} + \mathbb{E}_{\Pi_{\mathrm{e}}} B_r. \tag{4.5}$$

The first term vanishes as $h \downarrow 0$. The second vanishes as $r \downarrow 0$: the atomic origin measure has total mass one, and each of its collision vertices has a positive lifetime.

This gives $\mathbb{P}(U = X) = 0$ for every invariant selected location. If Λ_ω had an atom, measurably selecting a largest one would contradict this conclusion. Hence Λ_ω is diffuse, and (4.4) gives $\nu(\{0\}) = 0$. The conclusion is conditional diffuseness of the survivor law given the invariant sigma-field.

Finally, monotonicity transfers the conclusion to any growing finite resampling count. For fixed k, eventually $k_N \ge k$, so

$$\liminf_N p_N(k_N) \ge p(k).$$

Now send $k \to \infty$ in (1.1). For the active configuration at a fixed positive cutoff, adaptive removal similarly gives conditional diffuseness of the counting measure. The expected common active-label count has a nonincreasing finite resampling profile. Its fixed-k limit tends to zero as $k \to \infty$, so the same monotonicity argument applies to $k_N \to \infty$. Hence the two finite configurations become disjoint in probability under every growing number of coordinate resamplings, at each fixed positive cutoff.

Full argument: [W, Sections 3, 7.6, 7.8–7.10, 7.13–7.18, and 7.28].

5 The mean history cost

5.1 The mass captured by one collision

For a background edge e, let W_e be the Lebesgue measure of insertion origins that destroy that edge. Conditional on the background, the expected number of its edges destroyed by one uniform insertion is $\sum_e W_e$. The discrepancy path alternates between lost background edges and new edges. Apart from an endpoint correction, its total length is twice the number of lost background edges.

There is a useful locality bound for W_e. The two neighboring gaps and the collision gap merge into a parent interval I_e of length L_e. An origin outside that interval cannot destroy e, so $W_e \le L_e$. At active count m, cyclic gap exchangeability gives

$$\mathbb{E}\sum_i x_i^2 \le C/m, \qquad \mathbb{E}L_e^2 \le C/m^2. \tag{5.1}$$

For example, conditional on the unordered gaps and their minimum s, the two neighbors of s are a uniform ordered pair of the other gaps. Expanding the increase of the gap-square sum under their merger gives an expected increment at most C/m^2. Summing over successive counts proves the first estimate, and another expansion gives the second.

For two independent insertions, an edge destroyed by both has conditional probability W_e^2. By (5.1), $\mathbb{E}\sum_e W_e^2$ is bounded uniformly in the initial size. The overlap of the two discrepancy paths therefore contributes only a bounded error to the expected resampling distance:

$$\mathbb{E}H_N(1) = 4\mathbb{E}\sum_e W_e + O(1). \tag{5.2}$$

The factor four is two paths, each counting both lost and new edges.

To evaluate the sum, let $s_{M,m}$ be the next collision separation in a background of initial size M, just before its active count falls from m to $m-2$. Each gap of length x_i carries its pushed origin measure μ_i. Define its two corner regions to have widths $\min(s_{M,m}, x_i/2)$, and let $A_{M,m}$ be their total mass over all gaps. The central collision gap contributes all its mass $s_{M,m}$. The two neighboring gaps contribute their corner masses. Exchangeability of the *decorated* gaps therefore gives the exact expectation identity

$$\mathbb{E}W_{M,m} = \mathbb{E}s_{M,m} + \frac{\mathbb{E}A_{M,m} - \mathbb{E}s_{M,m}}{m-1}. \tag{5.3}$$

The decoration is essential: the expected origin measure in a gap is not simply Lebesgue measure.

The two inputs to (5.3) are

$$ms_{M,m} \longrightarrow c, \qquad A_{M,m} \longrightarrow \tfrac{1}{2} \quad \text{in } L^1, \qquad m \to \infty, \quad m/M \to 0, \tag{5.4}$$

where c is defined in (3.2). The first is inversion of the renewal count law. To understand the second, consider a deterministic cutoff r instead. An insertion changes the active count by -1 when its image lies in a corner and by $+1$ otherwise. Hence, with $K_M(r) = |C_r(B_M)|$,

$$\mathbb{E}A_M(r) = \frac{1 - \{\mathbb{E}K_{M+1}(r) - \mathbb{E}K_M(r)\}}{2}. \tag{5.5}$$

At a mesoscopic cutoff, the leading count is c/r for either initial parity, and the exact renewal formulas make the difference on the right tend to zero. Independent marked gaps give a renewal-reward law and therefore concentration of the corner mass around $1/2$.

Transferring this statement uniformly to fixed initial count takes some care. For a Poisson background the marked first and second moments give the law throughout $r \to 0$, $Mr \to \infty$. Coupling Poisson and fixed counts changes its corner mass in mean by at most $C\sqrt{M}\, r$, which covers small cutoffs. Finite-tree coefficient estimates cover the complementary larger cutoffs. A separate bracket around the random collision separation then proves (5.4). Substituting a random time directly into (5.5) would not be justified.

Now sum (5.3) over $j > m$ of the appropriate parity. The active count decreases by two, so the harmonic sum is $\frac{1}{2}\log(M/m) + O(1)$. Equations (5.4) give

$$\mathbb{E} \sum_{j>m} W_{M,j} \sim a_{\mathrm{H}} \log(M/m), \qquad 2 \le m = o(M), \tag{5.6}$$

where the sum uses the background parity and

$$a_{\mathrm{H}} = \frac{c + 1/2}{2} = \frac{1 + e^{-\gamma}}{4}.$$

A central range of counts gives the asymptotic; deterministic bounds $j s_{M,j} \le 1$ and $A_{M,j} \le 1$ control its omitted ends. Equation (5.2) now explains the coefficient in Theorem 1.2.

5.2 Stopping at the edit scale

For many edits, use an iid uniform background B of size M and q independent uniform probes, independent of B and divided into two sets P_1, P_2. Resampling corresponds to $M = N - k$ and $q = 2k$. Choose $2q + 4 \le m < M$ with $m \equiv M \pmod 2$, stop the background when its active count reaches m, and call that separation τ. Let $G_B(z) = |\mathcal{M}_\tau(B) \triangle \mathcal{M}_\tau(B + z)|$ be one probe's early cost. The useful comparison is

$$\mathbb{E}\left| H(B + P_1, B + P_2) - \sum_{i=1}^{q} G_B(Z_i) \right| \le C\left(m + q + \frac{q^2}{m - q} \right). \tag{5.7}$$

Here $H(A, A') = |\mathcal{M}(A) \triangle \mathcal{M}(A')|$, and Z_i lists all the probes.

The terms have a concrete meaning. After τ, discrepancy contraction leaves only $O(m + q)$ possible further edges. Before τ, telescope the effects of the probes. A failure of additivity has a mixed two-probe witness: in one of the four configurations with neither, either, or both probes present, a parent interval contains both sources. Summing squared parent masses over collisions above count $m - q$ costs $C/(m - q)$. There are order q^2 possible pairs. The argument uses the appropriate positive configuration for each witness; it does not place every witness in the original background. That distinction allows boundary cascades.

For $q = o(M)$, choose the parity-compatible m in $4q + 4 \le m \le 4q + 5$. Then the error in (5.7) is $O(q)$. Conditional on B,

$$\mathbb{E}[G_B(Z) \mid B] = 2 \sum_{j>m} W_{M,j} + O(1).$$

Combining with (5.6) gives

$$\mathbb{E} H(B + P_1, B + P_2) \sim 2q a_{\mathrm{H}} \log(M/q).$$

For $q = 2k$ and $M = N - k$, this is $(1 + e^{-\gamma}) k \log(N/k)$. The range below the edit spacing produces the logarithm; the interacting late part costs only order k in mean.

Full argument: [W, Sections 7.11–7.12 and 7.19–7.22].

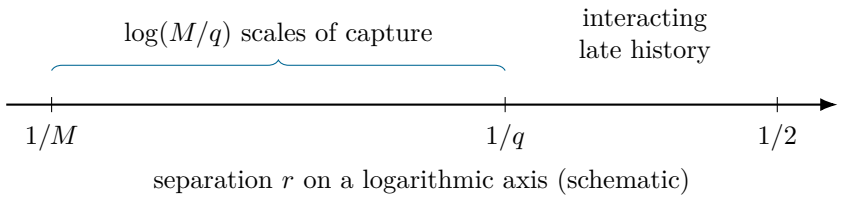


Figure 2: With q sparse probes, most of the history cost accumulates between the input scale and the edit scale. The later part has mean order q and centered fluctuations of order $\sqrt{q}$.

6 Concentration of the history cost

The fixed-edit and growing-edit laws use different averaging mechanisms. When there are many probes, average over origins in one background. When there are only finitely many, average along logarithmic scales. Both use the capture representation, with the averaging variable chosen to match the number of edits.

For growing $q = o(M)$, the captures $G_B(Z_i)$ in (5.7) are conditionally iid given B. A single insertion has a second moment bounded by $C(1 + \log M)^2$, so

$$\mathbb{E}\left|\sum_{i=1}^{q} G_B(Z_i) - q\mathbb{E}[G_B(Z) \mid B]\right| \le C\sqrt{q}(1 + \log M). \tag{6.1}$$

This is negligible compared with $q \log(M/q)$ for every growing $q = o(M)$. For $q \le \sqrt{M}$ use $\log(M/q) \ge \frac{1}{2}\log M$; for $q > \sqrt{M}$ the factor $q^{-1/2}$ controls the remaining logarithm.

The background conditional mean must also concentrate. Reveal the cyclic order of decorated gaps only as mergers require it: at each step expose the two neighbors of the minimum and contract that three-gap block. The unexposed cyclic order remains uniform, even given the entire revealed history. In this filtration,

$$W_{M,j} - \left(s_{M,j} + \frac{A_{M,j} - s_{M,j}}{j-1}\right)$$

is a martingale difference. The sum above active count m has variance at most $\sum_{j>m} \mathbb{E}L_{M,j}^2 \le C/m$. The predictable part is controlled in L^1 by (5.4) and the harmonic-sum argument. Consequently

$$\frac{\sum_{j>m} W_{M,j}}{\log(M/m)} \longrightarrow a_{\mathrm{H}} \quad \text{in } L^1.$$

Together with (6.1) and the $O(q)$ comparison error, this proves the growing-edit part of Theorem 1.2.

For fixed q, take the stationary marked line entrance process and insert one origin at zero. Let $J(a, b)$ count background edges with separations in $(a, b]$ destroyed by this insertion. Scaling the entrance law makes the costs in successive unit logarithmic bands stationary, with

$$\mathbb{E}J(a, b) = a_{\mathrm{H}} \log(b/a). \tag{6.2}$$

The marked renewal process is translation-ergodic. This also forces ergodicity of its dilation by e: conjugating a fixed translation by repeated dilations makes that translation arbitrarily small. An L^2 function invariant under dilation is therefore invariant under every translation, and hence constant. The ergodic theorem gives

$$\frac{J(e^{-T}, 1)}{T} \longrightarrow a_{\mathrm{H}} \quad \text{in } L^1.$$

The averaging here is over spatial dilations of the line process.

There is still a finite-sample transfer to make. Keep a central range of separations $M^{-1+\epsilon} \le r \le M^{-\epsilon}$, cap each unit-band reward, and group a fixed number of bands into a block. Marked local convergence compares each finite block with its line counterpart, uniformly over the central range. The triangle inequality then compares the average of all blocks without requiring their independence. First let $M \to \infty$, then increase the block size using line ergodicity. Finally remove the cap and let $\epsilon \downarrow 0$.

The last removal uses positivity: the capped central capture is bounded by the full capture, whose mean is already known sharply. Thus the discarded tails cannot hide a macroscopic part of the normalized cost. A fixed number of probes has only a bounded expected interaction error, so the same law holds for every fixed q. Combining fixed and growing subsequences proves (1.2) for all nonzero $k = o(N)$. For label histories, the spatial and label distances differ by at most $2k$, which is negligible on this scale. At positive edit density, independent permutations of the refreshed labels give the stronger uniform mean bound $\mathbb{E}(H_N - H_N^{\mathrm{lab}}) \le 4$.

Full argument: [W, Sections 7.24–7.25].

7 Fluctuations along the resampling path

The two main theorems describe the survivor response and the typical history cost. Fluctuations require finer control of the same discrepancy paths: their correlations across scales, their interactions, and the part of the history after those interactions become frequent. We describe the additional estimates separately in the sparse, dense, and near-complete resampling regimes.

7.1 Sparse edits: captures and their remainder

For the single line insertion, logarithmic capture bands satisfy a functional central limit theorem. Their covariance decays sufficiently fast, and

$$\frac{\operatorname{Var} J(e^{-T}, 1)}{T} \longrightarrow v_{\mathrm{H}} > 0.$$

The mechanism combines the origin-map approximation with marked renewal blocks separated by long gaps. Dilation turns spatial separation estimates into logarithmic-time decorrelation. Moment factorization controls higher cumulants and yields Brownian fluctuations. A separate argument rules out zero variance; v_{H} has not been explicitly evaluated.

For q probes, twice the sum of their captures suggests variance $4qv_{\mathrm{H}} \log(M/q)$. The $O(q)$ mean error in (5.7) can exceed the proposed standard deviation. To obtain a central limit theorem, we center the interaction remainder and prove that its variance is $O(q)$. The late history needs the same bound.

For the early remainder, an empty background cutoff gap separates the histories of all probe subsets. Consecutive occupied gaps form clusters. Their rewards include cascades across adjacent gaps, which can be arbitrarily long in a deterministic configuration. A fine-scale interaction is rare, however, and its probability supplies a summable gain in each logarithmic band. The cluster decomposition accommodates the changing boundaries and produces uniform block moments and the $O(q)$ variance bound.

To impose the prescribed input count, expose complete original records only inside probe-occupied gaps. The unused gaps retain random original counts. Those hidden counts form a reservoir whose local central limit estimate absorbs the conditioning on the prescribed total. Separate moment bounds transfer centers and variances; total variation convergence alone would not transfer them.

Late edges use marginal parent locality. With high probability, nearby fully shielded intervals determine a cell's history against arbitrary changes farther away. Their failure probability decreases exponentially with the search radius. Summing localized covariances over dyadic separation bands gives centered late variance $O(q)$. This remains valid when the two copies have few common active points.

For k resamplings, the complete sparse result is consequently

$$\frac{H_N(k) - \mathbb{E}H_N(k)}{\sqrt{8kv_{\mathrm{H}}\log(N/k)}} \Longrightarrow \mathcal{N}(0,1), \qquad 1 \le k = o(N), \tag{7.1}$$

with the corresponding variance asymptotic. Bounded counts use a joint comparison with finitely many independent line captures; growing counts use the centered remainder and reservoir estimates.

Uniform versions resolve two parameters at once. For $k \to \infty$, $k = o(N)$, put $M = N - k$, $q = 2k$, $L = \log(M/q)$ and

$$r_N(t) = \begin{cases} M^{-1}(M/q)^t, & 0 \le t \le 1, \\ q^{-1}(q/2)^{t-1}, & 1 \le t \le 2. \end{cases}$$

After centering by its exact mean and dividing by $\sqrt{8kv_{\mathrm{H}}L}$, the field $H_N(\lfloor ku \rfloor; r_N(t))$ converges in $C([0,1] \times [0,2])$, after bilinear interpolation, to

$$W(u, t \wedge 1), \qquad \mathrm{Cov}(W(u,s), W(v,t)) = \min(u,v)\min(s,t). \tag{7.2}$$

The interpolation uses edit mesh $1/k$ and time mesh $1/\lceil L \rceil$; the raw field is uniformly close to it. Shared edit prefixes explain $\min(u,v)$; shared logarithmic bands explain $\min(s,t)$. Uniform centered control of the late history explains why the time coordinate stops at one. Tightness requires maximal bounds over both parameters; finite collections of capture limits alone do not give the sheet theorem.

Full argument: [W, Sections 7.26, 7.37–7.42, and 7.48–7.56].

7.2 Positive density: a profile and a conditioned Gaussian process

At edit fraction θ, take independent Poisson inputs on the line with common intensity $1 - \theta$ and private intensities θ, θ. Define $\mathfrak{h}(\theta)$ as the expected changed-edge density, assigning each edge to its left endpoint. At a fixed separation cutoff b, gaps longer than b in the union input separate iid blocks. A renewal-reward law gives the truncated density. The unprocessed edge density tends to zero as $b \to \infty$, yielding

$$H_N(k)/N \longrightarrow \mathfrak{h}(\theta) \quad \text{in } L^2 \quad \text{when } k/N \to \theta.$$

The function is continuous, increasing and concave, with $\mathfrak{h}(0) = 0$, $\mathfrak{h}(1) = 1$, and

$$\mathfrak{h}(\theta) \sim (1 + e^{-\gamma})\theta\log(1/\theta) \quad (\theta \downarrow 0). \tag{7.3}$$

This dense law, combined with the sparse laws, gives the uniform relative concentration stated in Theorem 1.2.

For fluctuations, the Poisson block covariance must be corrected because the finite model has prescribed counts of each input type. Let R be a block's vector of rewards, C its vector of color counts, and L_b its length. Center each by its mean rate times L_b and write the covariance per unit length as

$$\begin{pmatrix} V_b & B_b \\ B_b^{\mathsf{T}} & D \end{pmatrix}.$$

Here D is the diagonal matrix of color intensities, with zero-intensity colors omitted. Conditioning the counts gives the reward covariance

$$\Gamma_b = V_b - B_b D^{-1} B_b^{\mathsf{T}}. \tag{7.4}$$

The subtraction removes fluctuations caused merely by random input counts. Local count reservoirs justify this conditioning, and late locality removes b.

For the whole nested resampling path, colors specify intervals of old and fresh priorities. Rare changes of priority give increment moment bounds at fixed b. A further uniform late bound is

$$\Big\| \max_{0\le j\le N} \sup_{r\ge b/N} |H_N(j;r) - H_N(j;b/N) - \mathbb{E}\{H_N(j;r) - H_N(j;b/N)\}| \Big\|_p \le C_p \sqrt{N/b}\log(2+b),$$

where the separation supremum ends at $1/2$. This makes the cutoff removable simultaneously for all edit fractions. Together with a uniform mean bias $|\mathbb{E}H_N(j) - N\mathfrak{h}(j/N)| \le C\log^2(N+2)$, it gives, after linear interpolation, convergence in $C([0,1])$:

$$\left(\frac{H_N(\lfloor N\theta\rfloor) - N\mathfrak{h}(\theta)}{\sqrt{N}}\right)_{0\le\theta\le1} \Longrightarrow (G(\theta))_{0\le\theta\le1}. \tag{7.5}$$

The limit is centered, continuous and Gaussian, with covariance $\Gamma = \lim_{b\to\infty}\Gamma_b$. It has $G(0) = G(1) = 0$ and $\operatorname{Var} G(\theta) = v_{\mathrm{D}}(\theta) > 0$ for $0 < \theta < 1$.

The count contribution in (7.4) can be expressed through the mean profile. Let $v_{\mathrm{P}}(\theta)$ be the unconditioned Poisson variance per unit length. Then

$$\begin{aligned} v_{\mathrm{P}}(\theta) - v_{\mathrm{D}}(\theta) &= (1-\theta)\big(\mathfrak{h}(\theta) - \theta\mathfrak{h}'(\theta)\big)^2 \\ &\quad + \frac{\theta}{2}\big(\mathfrak{h}(\theta) + (1-\theta)\mathfrak{h}'(\theta)\big)^2. \end{aligned} \tag{7.6}$$

Poisson count differentiation identifies reward–count covariances with intensity derivatives of the mean. Homogeneity and symmetry between the two private colors determine these derivatives from $\mathfrak{h}$ and $\mathfrak{h}'$; uniform late variance control justifies passage to the completed history. The remaining conditioned covariance Γ is still implicit.

Both functional limits transfer to label histories. Put $D_j(r) = H_N(j;r) - H_N^{\mathrm{lab}}(j;r)$. Label coincidences accumulate, so $0 \le D_j(r) \le D_j(1/2)$. The uniform bound $\mathbb{E}D_j(1/2)^2 \le 32$ and a union bound over $j \le k$ make the correction negligible on the sparse sheet scale. For the dense path, the stronger estimate

$$\Big\| \max_{0\le j\le N} D_j(1/2) \Big\|_p \le C_p \log(N+2)$$

makes it negligible relative to $\sqrt{N}$.

The mean comparison needs sharper forms near the boundaries. They give the common terminal normalization

$$\frac{H_N(k) - N\mathfrak{h}(k/N)}{\sqrt{N v_{\mathrm{D}}(k/N)}} \Longrightarrow \mathcal{N}(0,1) \quad \text{whenever } 1 \le k \le N, \quad (N-k)^2/N \to \infty. \tag{7.7}$$

Here $\operatorname{Var} H_N(k) \sim N v_{\mathrm{D}}(k/N)$, and $v_{\mathrm{D}}(\theta) \sim 8 v_{\mathrm{H}}\theta\log(1/\theta)$ at zero. Replacing the center by only the leading expression in (7.3) is not justified at the fluctuation scale.

Full argument: [W, Sections 7.23, 7.43, 7.46–7.57, and 7.61].

7.3 Near complete resampling: rare shared edges

Let $M = N - k$ be the number of unchanged coordinates, now small relative to N. If J_N is the number of common spatial edges, then exactly

$$H_N(k) = N - 1 - 2J_N. \tag{7.8}$$

A common edge needs two unchanged endpoints. Among pairs at distance of order $1/N$, the number of opportunities is therefore of order M^2/N. Conditional on a candidate pair, the two surrounding private inputs become asymptotically independent.

Let $a(t)$ be the probability that two forced points at distance t are matched in a unit-rate Poisson input on the line, and set

$$\kappa = \int_0^\infty a(t)^2\, dt.$$

For bounded rescaled lengths, disjoint-pair factorial moments give the Poisson law; locality supplies the two acceptance factors $a(t)$. A shared pair of distinct matching edges cannot reuse an endpoint, which controls the remaining factorial-moment terms. Tail estimates then remove the length cutoff. Thus

$$M^2/N \to \lambda < \infty \quad \Longrightarrow \quad J_N \Longrightarrow \mathrm{Pois}(\kappa\lambda). \tag{7.9}$$

There is also a marked point-process version, with intensity $\lambda\, dx\, a(t)^2\, dt$ for edge location x and rescaled length t. The coefficient has the certified enclosure

$$0.194237478218 \le \kappa \le 0.194237478220.$$

For label edges there is an additional asymptotically independent Pois(1/2) coincidence count, so the bounded endpoint law depends on which edge convention is used.

When $M = o(N)$ and $M^2/N \to \infty$, the shared-edge count is instead asymptotically Gaussian, with mean and variance asymptotic to $\kappa M^2/N$. This matches (7.7) because

$$v_{\mathrm{D}}(\theta) \sim 4\kappa(1-\theta)^2 \quad (\theta \uparrow 1).$$

The same constant appears in the separate integer matching spectrum

$$1 - \mathfrak{h}(\theta) = \sum_{\ell \ge 2} \pi_\ell (1-\theta)^\ell, \qquad \pi_2 = 2\kappa.$$

The weights sum to one. They are limits of the normalized orthogonal weights of the labeled matching indicators, kept at integer coordinate orders; the survivor spectrum ν instead rescales those orders by N. A Poisson difference-kernel representation writes each π_ℓ as an integral of a square. Geometrically separated inserted points give a positive contribution, proving $\pi_\ell > 0$ for every $\ell \ge 2$. Consequently every fixed truncation of this series has a nonzero next term. For fixed $J \ge 2$, in the Gaussian endpoint regime, centering by the degree-J truncation is valid precisely when $M = o(N^{1-1/(2J)})$; at the critical scale the omitted term produces a nonzero shift.

The endpoint in (7.9) is complete *resampling*, not the last physical collision. For the latter, if R_N is the last collision separation of an odd sample, the working result is

$$R_N \Longrightarrow \frac{1}{1 + Y_1 + Y_2},$$

where Y_1, Y_2 are iid with density f from (3.1). Its physical time is half this quantity. This constrained renewal law supplies no assertion of Poisson terminal clumps. The Poisson mechanism above concerns rare pairs shared by almost independently resampled inputs.

Full argument: [W, Sections 7.7, 7.23, 7.44–7.47, and 7.59].

7.4 The two boundary limits of the Gaussian process

The sparse and dense path theorems describe the same finite resampling path at different scales. Their connection identifies the behavior of G near zero. As $\delta \downarrow 0$,

$$\left(\frac{G(\delta u)}{\sqrt{8 v_{\mathrm{H}} \delta \log(1/\delta)}}\right)_{0 \le u \le 1} \Longrightarrow (B(u))_{0 \le u \le 1} \quad \text{in } C([0,1]), \tag{7.10}$$

where B is standard Brownian motion. To join the two limits, use the profile-centered version of the sparse path theorem: at the grid $u = j/k$, its numerator is exactly $H_N(j) - N\mathfrak{h}(j/N)$, and its linear interpolation agrees with the restriction of the dense path. Given any $\delta_n \downarrow 0$, choose N_n large enough that the dense law approximates $G(\delta_n \cdot)$ at its rescaled size and $k_n = \lfloor N_n \delta_n \rfloor \to \infty$. Since $k_n = o(N_n)$, the sparse theorem applies along this same sequence and gives (7.10). The use of an arbitrary sublinear sequence is what makes the diagonal choice legitimate.

Near one, the shared-edge description gives

$$\frac{\Gamma(1-\delta u, 1-\delta v)}{2\pi_2 \delta^2} \longrightarrow \min(u,v)^2 \quad \text{uniformly on } [0,1]^2. \tag{7.11}$$

At a fixed separation cutoff, the same edge appearing at two retained fractions requires two old tags; distinct edges require four. This leaves the leading covariance proportional to $\min(u,v)^2$, with a count-conditioning correction of order δ^3. A diagonal selection applies the tagged late-variance bound to the dense covariance and removes the cutoff. Gaussianity and finite-rank approximation then give

$$\left(\frac{G(1-\delta u)}{\sqrt{2\pi_2}\,\delta}\right)_{0 \le u \le 1} \Longrightarrow (B(u^2))_{0 \le u \le 1}$$

in finite-dimensional distributions and in $L^2([0,1])$. Convergence in $C([0,1])$ remains open: it requires a uniform modulus estimate in addition to (7.11).

Full argument: [W, Sections 7.58 and 7.60].

8 What the arguments leave open

The survivor theorem is qualitative at its sharp scale: it identifies when agreement vanishes, but does not identify ν explicitly. The working arguments give logarithmic upper bounds on agreement and power bounds for finite-sample approximation, with uncalibrated constants. A narrow certified interval for p_∞ remains a separate numerical and analytic problem. The historical estimate near 0.658 and the original small-sample calculations still lack their complete original reproduction package.

The history laws reveal more explicit structure. Their leading sparse coefficient comes from the renewal density and the corner-mass identity; their complete-refresh coefficient comes from the squared forced-pair acceptance probability. Between these endpoints, the profile $\mathfrak{h}$, the variance rate v_{H}, and the conditioned covariance of G remain largely implicit. The boundary limits isolate a more focused path question: prove tightness at the complete-resampling endpoint in the uniform topology. Ergodicity of the full circle refresh chain is also open.

The proof sketches also identify where a detailed verification should concentrate. The marked construction must preserve identities and origins, the adaptive insertion must preserve the stationary law, and the corner law must be uniform through the required finite-sample regimes. For fluctuations, the centered remainder estimates and the exact-count conditioning are additional substantive inputs.

Finite checks of collapse maps, parent witnesses, and label coincidences support those constructions, but do not prove their asymptotic bounds.

The full record also treats spatial active-count fields and fluctuations of weighted parent histories. Those observables develop the same renewal machinery. Here the organizing object has been the origin of a perturbation: its discrepancy moves through the matching, the merger tree records that movement, and the accumulated captures determine both the final survivor response and the changed collision history.

Formal and computational companions. The versioned proof and verification map separates the full working proofs cited here from the exact interval computations and the finite Lean development. The latter verifies 193 public theorems through `Stable.lean`; it does not formalize the entrance-law, noise, history or fluctuation limits. The numerical estimate near 0.658 remains uncertified.

References

[W] *The balloon process on the circle.* AI-written working manuscript prepared with Jacob Richey, mathematical record through 20 October 2026, including Sections 7.1–7.61. Full mathematical proof record, with TeX source.

[1] O. Angel, G. Ray and Y. Spinka. *A tale of two balloons.* arXiv:2103.06847, 2021.

[2] A. E. Holroyd, R. Pemantle, Y. Peres and O. Schramm. *Poisson matching.* arXiv:0712.1867, 2007, revised 2008.

[3] A. J. Bray, B. Derrida and C. Godreche. *Non-trivial algebraic decay in a soluble model of coarsening.* Europhys. Lett. **27** (1994), 175–180. Authors' scan.

[4] A. Faggionato, C. Roberto and C. Toninelli. *Universality for one-dimensional hierarchical coalescence processes with double and triple merges.* Ann. Appl. Probab. **24** (2014), 476–525. arXiv:1112.4630.

[5] D. Ahlberg, M. Deijfen and M. Sfragara. *Mean field stable matchings.* arXiv:2406.04911, 2024.